

草稿



State

$$\nu_t = \frac{1}{t} \sum_{j=1}^t v_j \quad (1)$$

$$\kappa_t = \frac{1}{t} \sum_{j=1}^t k_j \quad (2)$$

$$D_t \triangleq \sum_{j=1}^t (v_j - \nu_t)(k_j - \kappa_t)^\top \quad (3)$$

Recurrence

$$\nu_t = \nu_{t-1} + \frac{1}{t}(v_t - \nu_{t-1}) \quad (4)$$

$$\kappa_t = \kappa_{t-1} + \frac{1}{t}(k_t - \kappa_{t-1}) \quad (5)$$

$$D_t = D_{t-1} + \frac{t-1}{t}(v_t - \nu_{t-1})(k_t - \kappa_{t-1})^\top \quad (6)$$

Sequence Parallel

$$P_t = \text{PrefixSum}(v)_t \quad (7)$$

$$R_t = \text{PrefixSum}(k)_t \quad (8)$$

$$M_t = \text{PrefixSum}(vk^\top)_t \quad (9)$$

$$D_t = M_t - \frac{1}{t} P_t R_t^\top \quad (10)$$

Output

$$\Delta q_t = q_t - q_{t-1} \quad (11)$$

$$\beta_t = \frac{\exp(k_t^\top q_t)}{(t-1) + R_{t-1}^\top q_t} \quad (12)$$

$$o_t = (1 - \beta_t)(o_{t-1} + D_t \Delta q_t) + \beta_t v_t \quad (13)$$

